MIL-S-24473B(SH)
19 December 1977
SUPERSEDING
MIL-S-24473A(SHIPS)
28 December 1973

MILITARY SPECIFICATION

SAW, CIRCULAR, ELECTRIC, PORTABLE, DOUBLE-INSULATED

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a double-insulated, portable, electric, circular saw.

1.2 Classification. The saw shall be of the following type and sizes, as specified (see 6.2).

Type I - Standard.
Size - 6 to 6-3/4 inch nominal blade diameter.
Size - 7 to 7-1/2 inch nominal blade diameter.
Size - 8 to 8-1/4 inch nominal blade diameter.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-P-40 - Packaging and Packing of Hand Tools.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1337 - General Suppression System Design Requirements For Portable Electric Hand Tools.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC.
Standards for Safety
UL-45 - Portable Electric Tools.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, Long Island, New York 11746; and 1655 Scott Boulevard, Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5130

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3.2 General requirements.

3.2.1 Design. The saw shall be designed for ease of operation and safety in use. The saw shall include all components, parts, and features necessary to meet the requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.2 Lubrication. All bearings (except sealed-for-life type) and mating gears shall be provided with a means of lubrication. Gears shall operate in a lubricant which shall be prevented from leakage from the gear casing or entering the motor enclosure. Parts requiring protection from dust shall be enclosed.

3.2.3 Interchangeability. All parts from the same manufacturer shall be manufactured to standards which will permit replacement or adjustment of similar parts without modification to parts or machine and insure ease of interchangeability.

3.2.4 Construction. The saw shall be constructed of parts which are new, without defects, and free of any characteristics that will prevent the saw from passing any of the examinations or tests specified herein. The saw shall be complete so that when connected to the specified power source, it can be used for any operation for which it was designed.

3.2.5 Insulation. The saw shall be double-insulated as defined in 6.4 and shall be in accordance with the requirements of UL-45.

3.2.6 Castings and forgings. All castings and forgings shall be free from defects or mismatching. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used for reclaiming any parts of the saw.

3.2.7 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are included in the fabrication of the original design. Those techniques shall not be employed as repair measures for defective parts.

3.2.8 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with a means of preventing loss of tightness. All parts, subject to removal or adjustment, shall not be swaged, peened, staked, or permanently deformed.

3.2.9 Surfaces. Surfaces of castings, forgings, and molded parts shall be clean and free of sand, dirt, fins, scale, flux, flashing, or harmful and extraneous materials. External surfaces shall be smooth and all edges rounded or beveled. Surface finish shall be manufacturer's standard practice.

3.2.10 Identification plate. The saw shall be provided with a suitable nameplate permanently marked and securely fastened to the housing in a conspicuous location. The nameplate shall enumerate, as a minimum, the information specified in UL-45.

3.2.11 Workmanship. Workmanship of the saw shall be such that the tool shall be free of defects which may adversely affect the operation efficiency, serviceability, and life expectancy of the tool.

3.3 Detail requirements.

3.3.1 Conformance to standards of Underwriters' Laboratories, Inc. The contractor shall submit to the contracting agency proof that the basic saw he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories, Inc. Standards for Electric Tools (UL-45). The label or listing with re-examination of the UL may be accepted as evidence that the saw conforms to this requirement. In lieu of the label or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the saw conforms to the applicable requirements of the published standards including methods of tests of the UL Standards noted above. Compliance with requirements of UL regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his saw. In the event of difference between any of the requirements of this specification and those of UL, the requirements of this specification shall govern.

3.3.2 Electromagnetic interference (EMI) suppression. When specified (see 6.2), the saw shall comply with the (EMI) suppression requirements of MIL-STD-1337 and be furnished with a label, permanently marked and attached in a conspicuous location, stating:

"This tool is EMI suppressed".

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3.3.3 Sizes and capacities. The saw shall conform to the requirements of table I for the size specified. The saw shall be in accordance with figure 1, with the motor armature axis parallel with saw-blade-spindle axis.

TABLE I. Saw requirements.

Size	Blade diameter	Minimum no-load blade speed 1/	Cutting capacity		Rated current (Max)	Blade arbor hole diameter 2/
			90 degrees (Min)	45 degrees (Min)		
	Inches	RPM	Inches	Inches	Amps	Inch
6	6 to 6-3/4	5500	2	1-11/16	10	5/8 (0.625)
7	7 to 7-1/2	5200	2-3/8	1-13/16	11	5/8 (0.625)
8	8 to 8-1/4	5100	2-3/4	2- 3/16	12	5/8 (0.625)

1/ No-load speed tolerance ± 10 percent.
2/ Arbor hole tolerance $^{+\ 0.003}_{-\ 0.000}$

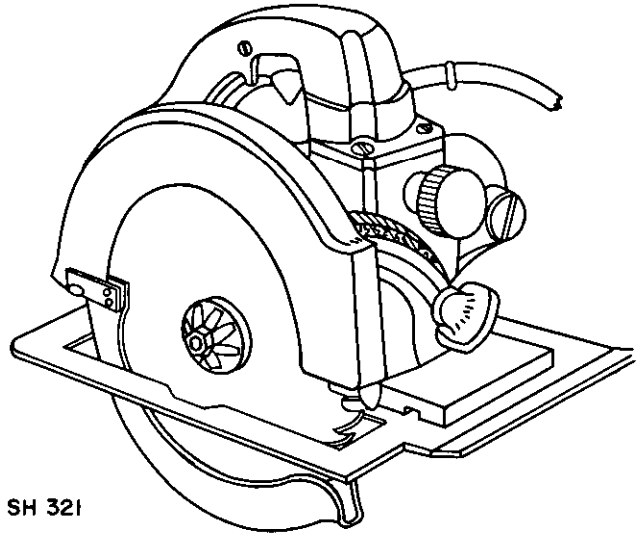


FIGURE 1. Portable, electric, circular saw, motor axis parallel with saw-blade-spindle axis.

3.3.4 Housing and handle(s). The saw housing and handle(s) shall be in accordance with the requirements of UL-45.

3.3.5 Motor. The saw motor and all working parts thereof shall be in accordance with the requirements of UL-45.

3.3.6 Wiring, cord, and plug cap. All wiring, flexible cord, and attachment end plug cap shall be in accordance with the requirements of UL-45. The length of cord with end plug cap shall be of the manufacturer's standard length offered.

3.3.7 Switch. The saw shall be provided with a single throw, instant release trigger switch integrally mounted in the saw handle and equipped with a device that automatically locks it in the "off" position when the actuator is released, so that two motions are required to energize the saw and preclude accidental starting.

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3.3.8 Bearings. All rotating parts of the saw shall be provided with precision quality, commercially available ball, roller, or caged needle bearings. The saw blade arbor bearing may be of the bronze sleeve oil impregnated type. Integral or separate shielding to prevent leakage of lubricant and entrance of contaminants shall be provided where necessary. Steel inserts shall be omitted in the mounting of bearing when sufficient contact area and fit between the bearing outer diameter and housing is provided. Bearings designed primarily for radial loads shall not be used to take thrust. Provision shall be made to absorb thrust loads.

3.3.9 Gears and shafts. Gears and shafts shall be manufactured from steels having physical properties and a hardness suitable for the requirements of the specific saw. All gears and shafts shall be supported by at least two bearings and shall be of such dimensions to prevent deflection. Gears shall have machine cut or formed teeth and be designed to efficiently transmit imposed loads. Gears and shafts shall be fully enclosed.

3.3.10 Base. Saws shall be provided with a metal base designed to support the saw against the material being cut and provided with adjustments for cutting depth and angle of cut. The base design shall include means for mounting a rip fence on either side.

3.3.10.1 Rip fence. Each saw shall be furnished with an adjustable rip fence to fit the base. The fence shall be graduated in increments of 1/4 inch or less and marked each inch to indicate a ripped width up to at least 5 inches. Graduation shall indicate distance inside of blade to edge of rip fence.

3.3.10.2 Position indicator. Provisions shall be made on the saw base to clearly indicate the point of saw blade contact with work at the 0-degree and 45-degree angle adjustment.

3.3.10.3 Angle adjustment. Provision shall be made for adjustment of the base from the 0-degree to 45-degree angle. A graduated angle scale shall clearly indicate the angle in increments of not more than 5 degrees from 0 through 45 degrees. Provisions shall be made for making angle adjustment without the use of tools.

3.3.10.4 Depth adjustment. The base shall be readily adjustable to control the depth of cut within the cutting capacity specified herein for the size specified. Base adjustment shall be accomplished without the use of special tools.

3.3.11 Dust blower. The saw shall be provided with a dust blower having an air output which shall be capable of clearing sawdust from the line of cut and away from the operator.

3.3.12 Saw blade. Each saw shall be furnished with one combination flatground saw blade of the contractor's standard commercial quality.

3.3.13 Instruction book. The manufacturer shall furnish one copy of his regular instruction book and parts list with each saw.

3.3.14 Carrying case. When specified (see 6.2) each saw shall be provided with a durable carrying case. Provisions shall be made for securely retaining the cover when closed. The case shall be of ample size, designed to hold the saw and accessories in a fixed position. The interior and exterior surfaces of the carrying case shall have a durable coating of paint or enamel, except surfaces which are inherently corrosion free or electroplated. The case shall be marked with the manufacturer's name or trademark.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 First article inspection. First article inspection shall consist of the examination of 4.5.1 and all tests specified in 4.6.

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4.3.1 Prior approval of EMI suppression. Prior to first article examination and tests, complete EMI suppression tests shall be performed to determine compliance with the requirements of 3.3.2 and a report of such tests made available to the government inspector.

4.4 Sampling for quality conformance.

4.4.1 Lot. All saws of the same size presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.4.2 Sample. A random sample shall be selected from each lot in conformance with MIL-STD-105 at inspection level II.

4.5 Quality conformance inspection. Quality conformance inspection shall be performed on each sample saw selected in accordance with 4.4.2. Quality conformance inspection shall consist of the following:

(a) Examination (see 4.5.1).
(b) Tests (see 4.6).
(c) Inspection of preparation for delivery (see 4.7).

4.5.1 Examination. Each saw selected in accordance with 4.4.2 shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table II, acceptable quality level to be as indicated. Nonconforming saws shall not be offered for delivery and if the nonconforming saws in any one sample exceed the acceptance number for that sample, the lot represented shall be rejected.

4.5.2 Tests for quality conformance. Each sample selected in accordance with 4.4.2 shall be tested for compliance with the requirements of this specification. The acceptable quality level (AQL) shall be 2.5 percent defective. Nonconforming saws shall not be offered for delivery and if the number of defective saws exceeds the acceptable number for the samples tested, the lot represented by the samples shall be rejected. Quality conformance tests shall consist of the tests specified (see 4.6).

TABLE II. Classification of defects.

Category	Defects
Critical	AQL 0.10 percent defective
001	Proof or certification of UL listing requirements prior to incorporation of EMI suppression.
Major	AQL 2.5 percent defective
101	Saws not of size specified.
102	Component parts missing, damaged, defective, or not as specified.
103	Switch defective.
104	Base adjustment not as specified.
105	Evidence of lubricant leakage.
106	Rip fence not graduated as specified.
107	Position indicator not as specified.
108	Angle adjustment not as specified or requires the use of tools; depth adjustment not as specified or requires the use of special tools.
109	Dust blower not provided.
110	Voltage and current not as specified.
111	Evidence of poor workmanship.
Minor	AQL 4 percent defective
201	Identification plate not as specified or missing.
202	Handbook or parts list missing.
203	Carrying case not as specified.
204	Saw blade missing or not as specified.

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4.6 Test procedure.

4.6.1 Motor tests. Each sample saw selected in accordance with 4.4.2 shall be tested to determine compliance with UL-45 for universal motors for the following:

(a) No-load speed.
(b) No-load current.
(c) Dielectric strength.

4.6.2 No-load tests. Each saw selected in accordance with 4.4.2 shall be operated at no-load for 15 minutes during which time it shall be checked for switch operation, indication of overheating, and noise and vibration.

4.6.3 Switch test. The switch shall conform to the requirements of UL-45 for contact pitting, burning, and mechanical failure.

4.6.4 Performance test. The saw shall be subjected to a load test for a period of at least 1 hour or until motor temperature has stabilized. The saw shall be tested for the following:

(a) Rated torque.
(b) Temperature rise at rated current.

At the conclusion of the test, the saw shall be examined for excessive wear and arc damage on the commutator.

4.7 Inspection of preservation-packaging, packing, and marking for shipment and storage. Preservation-packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5 or as specified in the contract.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2 (see 6.5).

5.1 Preservation-packaging, packing, and marking. Preservation-packaging, packing, and marking shall be in accordance with the levels and requirements of PPP-P-40. Levels of preservation-packaging and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The double-insulated electric saw covered by this specification is hand-guided, intended primarily for cutting wood products at angles from 0 to 45 degrees. Special saw blades allow ferrous and nonferrous metals and masonry and ceramic materials to be cut.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size saw required (see 1.2).
(c) Electromagnetic interference suppression, when required (see 3.3.2).
(d) Specify carrying case, when required (see 3.3.14).
(e) First article inspection and approval, if required (see 4.3).
(f) Level of preservation, packing, and packaging required (see 5.1).

6.3 First article inspection.

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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6.4 Double insulation. Double insulation, for the purpose of this specification, is defined as an insulation system comprised of (a) functional insulation and (b) protective insulation. These two insulations are physically separated and so arranged that they are not simultaneously subjected to the same deteriorating influences to the same degree to ensure protection against electrical shock.

6.5 Subcontracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5130-N262)

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